

### Problem B: Radio Channel Assignments

We seek to model the assignment of radio channels to a symmetric network of transmitter locations over a large planar area, so as to avoid interference. One basic approach is to partition the region into regular hexagons in a grid (honeycomb-style), as shown in Figure 1, where a transmitter is located at the center of each hexagon.

An interval of the frequency spectrum is to be allotted for transmitter frequencies. The interval will be divided into regularly spaced channels, which we represent by integers  $1, 2, 3, \dots$ . Each transmitter will be assigned one positive integer channel. The same channel can be used at many locations, provided that interference from nearby transmitters is avoided. Our goal is to minimize the width of the interval in the frequency spectrum that is needed to assign channels subject to some constraints. This is achieved with the concept of a span. The span is the minimum, over all assignments satisfying the constraints, of the largest channel used at any location. It is not required that every channel smaller than the span be used in an assignment that attains the span.

Let  $s$  be the length of a side of one of the hexagons. We concentrate on the case that there are two levels of interference

Requirement A: There are several constraints on frequency assignments. First, no two transmitters within distance of each other can be given the same channel. Second, due to spectral spreading, transmitters within distance  $2s$  of each other must not be given the same or adjacent channels: Their channels must differ by at least 2. Under these constraints, what can we say about the span in,

Requirement B: Repeat Requirement A, assuming the grid in the example spreads arbitrarily far in all directions.

Requirement C: Repeat Requirements A and B, except assume now more generally that channels for transmitters within distance  $k$  differ by at least some given integer  $k$ , while those at distance at most  $k$  must still differ by at least one. What can we say about the span and about efficient strategies for designing assignments, as a function of  $k$ ?

Requirement D: Consider generalizations of the problem, such as several levels of interference or irregular transmitter placements. What other factors may be

important to consider?

Requirement E: Write an article (no more than 2 pages) for the local newspaper explaining your findings.

### 问题 B: 无线电信道分配

我们寻找无线电信道配置模型.在一个大的平面区域上设置一个传送站的均衡网络,以避免干扰.一个基本的方法是将此区域分成正六边形的格子(蜂窝状),如图 1.传送站安置在每个正六边形的中心点.

容许频率波谱的一个区间作为各传送站的频率.将这一区间规则地分割成一些空间信道,用整数  $1, 2, 3, \dots$  来表示.每一个传送站将被配置一正整数信道.同一信道可以在许多局部地区使用,前提是相邻近的传送站不相互干扰.根据某些限制设定的信道需要一定的频率波谱,我们的目标是极小化频率波谱的这个区间宽度.这可以用跨度这一概念.跨度是某一个局部区域上使用的最大信道在一切满足限制的配置中的最小值.在一个获得一定跨度的配置中不要求小于跨度的每一信道都被使用.

令  $s$  为一个正六边形的一侧的长度.我们集中考虑存在两种干扰水平的一种情况.

要求 A: 频率配置有几个限制,第一,相互靠近的两个传送站不能配给同一信道.第二,由于波谱的传播,相互距离在  $2s$  内的传送站必须不配给相同或相邻的信道,它们至少差 2.在这些限制下,关于跨度能说什么.

要求 B: 假定前述图 1 中的格子在各方向延伸到任意远,回答要求 A.

要求 C: 在下述假定下,重复要求 A 和 B.更一般地假定相互靠近的传送站的信道至少差一个给定的整数  $k$ ,同时那些隔开一点的保持至少差 1.关于跨度和关于设计配置的有效策略作为  $k$  的一个函数能说点什么.

要求 D: 考虑问题的一般化,比如各种干扰水平,或不规则的传送站布局.其他什么因素在考虑

中是重要的.

要求 E: 写一篇短文(不超过两页)给地方报纸,阐述你的发现.